

JOSEPH FIORESI (PH.D. CANDIDATE)

📍 Orlando, FL ✉️ joefioresi718@gmail.com 📞 630-818-0625 🌐 GitHub 🌐 Webpage **in** LinkedIn 📄 Google Scholar
U.S. Citizen **Research Interests:** Privacy-Preserving Computer Vision, Responsible AI, Bias Mitigation,
Large Multimodal Models (video-language), Video Understanding, Agentic Systems

Education

Ph.D. in Computer Science University of Central Florida, USA Aug 2022 – May 2027
Advisor: Dr. Mubarak Shah, GPA: 4.0/4.0 ORC Doctoral Fellowship

B.Sc. in Computer Science University of Central Florida, USA 2018 – 2022
GPA: 4.0/4.0 Mathematics Minor / National Merit Finalist / Benacquisto Scholarship

Work Experience

Adobe Inc., Seattle, USA *Research Scientist/Engineer Intern* May 2026 – Present
↪ Pankaj Nathani, Dr. Fabian Caba Heilbron, Shwetha Ram

- Working on developing a novel self-learning agentic system for cross-video understanding.
- Filed for a patent, submitted paper currently under review.

Adobe Inc., Seattle, USA *Research Scientist/Engineer Intern* May 2025 – Nov 2025
↪ Pankaj Nathani, Dr. Fabian Caba Heilbron, Dr. Kushal Kafle

- Developed an efficient model to generate useful embedding space transformations via natural language inputs, achieving high clustering/retrieval performance while maintaining the benefits of a useful semantic embedding space.
- Worked closely with product teams and end users to validate and discuss implementation of our model.
- Filed for a patent, internship paper accepted to ECCV 2026.

Center for Research in Computer Vision (CRCV) *Graduate Research Assistant* Aug 2022 – Present

- Perform rigorous theoretical computer vision research in various multimodal video understanding tasks with a specialization in privacy constraints and AI fairness.
- Regularly engaged in collaborative research with peers and professors, fostering interdisciplinary cooperation.

Better Solar LLC *Co-founder and Chief Technology Officer* Apr 2022 – Jan 2024

- Secured startup funding through the DoE American-Made Solar Prize Round 5 competition.
- Responsible for the development and deployment of a SaaS web application on Google Cloud Platform.

Computer Vision NSF REU *Undergraduate Student Researcher* May 2020 – Aug 2020

- Participated in a fast-paced learning program designed to give realistic research experience.
- Developed code to semantically segment defects in solar panels with a custom confidence threshold.
- Oral presentation at Photovoltaic Specialists Conference, invited for publication in IEEE Journal of Photovoltaics.

Publications

- Joseph Fioresi**, Fabian Caba, Pankaj Nathani, Mubarak Shah, Kushal Kafle. Controlling Embedding Spaces with Text-Conditioned Transformations, European Conference on Computer Vision (**ECCV**), 2026.
- Joseph Fioresi**, Parth Parag Kulkarni, Ashmal Vayani, Song Wang, Mubarak Shah. Learning to Share: Selective Memory for Efficient Parallel Agentic Systems, International Conference on Machine Learning (**ICML**), 2026.
- Swetha Sirnam, Rohit Gupta, Parth Parag Kulkarni, David G Shatwell, Jeffrey A Chan Santiago, Nyle Siddiqui, **Joseph Fioresi**, Mubarak Shah. VRR-QA: Visual Relational Reasoning in Videos Beyond Explicit Cues, IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- Joseph Fioresi**, Ishan Dave, and Mubarak Shah. Privacy Beyond Pixels: Latent Anonymization for Privacy-Preserving Video Understanding, International Conference on Learning Representations (**ICLR**), 2026.
- Adeel Yousaf, **Joseph Fioresi**, James Beetham, Amrit Singh Bedi, Mubarak Shah. SafeR-CLIP: Mitigating NSFW Content in Vision-Language Models While Preserving Pre-Trained Knowledge, AAAI Conference on Artificial Intelligence (**AAAI**), 2026.
- Joseph Fioresi**, Ishan Dave, and Mubarak Shah. ALBAR: Adversarial Learning approach to mitigate Biases in Action Recognition, International Conference on Learning Representations (**ICLR**), 2025.
- Joseph Fioresi**, Ishan Dave, and Mubarak Shah. TeD-SPAD: Temporal Distinctiveness for Self-supervised Privacy-preservation for video Anomaly Detection, IEEE/CVF International Conference on Computer Vision (**ICCV**), 2023.
- Joseph Fioresi**, Dylan Colvin, Rafaela Frota, Rohit Gupta, Mengjie Li, Hubert Seigneur, Shruti Vyas, Sofia Oliveira, Mubarak Shah, Kristopher O. Davis. Automated Defect Detection and Localization in Photovoltaic Cells Using Semantic Segmentation of Electroluminescence Images, IEEE Journal of Photovoltaics (**JPV**)/Photovoltaics Specialists Conference (PVSC), 2021.

9. (Under Review) Ashmal Vayani, Parth Parag Kulkarni, **Joseph Fiorese**, Mubarak Shah. MedRoute: RL-Based Dynamic Specialist Routing in Multi-Agent Medical Diagnosis, 2026.

Major Research Projects

Agentic Frameworks

Oct 2025 – present

- *Parallel Multi-agent Frameworks* (Oct 2025 - Apr 2026): We introduce an inter-team shared memory system for parallel agent teams working on web assistant tasks, showing improved performance at drastically reduced runtimes. The shared memory is controlled by a memory controller trained via our novel reinforcement learning setup.
- *Medical Specialist Agentic Framework* (Oct 2025 - Present): We design a flexible medical agentic framework that dynamically proposes and chooses relevant medical specialists to help solve each medical case. Our work demonstrates that the specialists used and the order in which they are chosen matter, and we use reinforcement learning to train a general practitioner agent to sequentially route the information between roles.

Large Multimodal Models (LMMs)/ Large Language Models (LLMs)

Mar 2024 – present

- *Benchmarking Implicit Video Understanding* (Mar 2025 - Nov 2025): We identified a shortcoming in existing evaluation of video models where many answers can be explicitly localized to individual frames. Instead, we carefully annotate videos where the question requires implicit reasoning between frames to solve, benchmarking a suite of SOTA models.
- *VQA Grounding in Video LMMs* (May 2024 - Sept 2024): Developed a framework to enable object-level VQA grounding in open-source LMM models (LLaVA-OneVision, VideoGPT+), improving the visual perception ability of LMMs.

Multimodal Vision-Language Models

Mar 2025 – Mar 2026

- *Embedding Space Personalization/Transformation* (May 2025 - Present): Trained a lightweight hypernetwork to take in natural language and output an affine transformation on multimodal embedding spaces to better emphasize the input attribute for attribute-aware retrieval from a frozen gallery.
- *Mitigating Unsafe Content in Multimodal Models* (Feb 2025 - July 2025): Identifying and mitigating NSFW content understanding in CLIP/SigLIP-based multimodal embedding models via robust unlearning techniques.

Privacy Preserving/Fair Video Understanding

Aug 2022 – present

- *Privacy Preserving General Video Foundation Models*[\[4\]](#) (Aug 2023 - Sept 2025): Developed a novel privacy preserving framework that moves anonymization training from the input pixel space to the latent feature space. The framework supports training across multiple downstream video understanding tasks, and shows state-of-the-art tradeoffs between privacy and utility across all tasks, including those unseen during anonymization training.
- *Video Debiasing Framework*[\[6\]](#) (Mar 2023 - Sept 2024): Developed a novel adversarial debiasing framework that reduces the effect of static biases (from both background and foreground) in video action recognition models. Without requiring additional labels/classifiers (scene, clothing, etc.), our method achieved SOTA on difficult bias evaluation protocols.
- *Privacy Preserving Video Anomaly Detection*[\[7\]](#) (Aug 2022 - March 2023): Developed a privacy-aware video anomaly detection framework utilizing temporally-distinctive video representations, achieving state-of-the-art tradeoff between privacy protection and utility performance on three popular weakly supervised VAD datasets.

Photovoltaic Defect Analysis

May 2020 – Dec 2021

- *PV Defect Segmentation Model/Dataset*[\[8\]](#) (May 2020 - Dec 2021): Helped plan out a project to automatically identify defects in electroluminescence images of solar modules. Annotated a dataset by hand with pixel-level defect masks labelled by category, developed model to achieve high segmentation performance.

Funded Projects

NSF Center for Smart Streetscapes (CS3) Engineering Research Center

Jan 2023 – Aug 2025

- Worked on projects in a multidisciplinary, cross-university engineering research center focused on developing smart city technologies to deploy in various urban environments. Focusing on situational awareness and privacy aspects.
- Served a year as a University Representative on the Student Leadership Council (SLC) to serve as a liaison between student and faculty members. Responsible for hosting numerous events and performing a center-wide SWOT analysis.
- Presented work to Harlem community members for feedback on real-world applicability of our work.

NSF Research Experience for Undergrad (REU)

2024

- Mentored undergraduate student in the program, led to 2nd place finish in an ECCV workshop challenge.

Awards and Honors _____

Frank M. Hubbard Engineering Endowed Scholarship Awardee	2025
2 nd place Google DeepMind Perception Test Challenge (Grounded VQA) (ECCV)	2024
eBay HSI Battle of the Brains Scholarship Awardee One of 4 chosen among 200+ competitors	2023
3 rd place HSI Battle of the Brains Competition	2023
ORCGS Doctoral Fellowship	2022-2027
National Merit Finalist	2018

Skills _____

Programming Languages	Python, MATLAB
Deep Learning frameworks	PyTorch, Keras
Tools/Frameworks	GCP, OpenCV, SciKit, Matplotlib

Coursework _____

- Advanced Computer Vision (*CAP 6412*)
- Computer Vision Systems (*CAP 6411*)
- Deep Learning Mathematics (*MAP 6197*)

Professional Services _____

- Reviewer for CVPR, NeurIPS, ICLR, ICML, ICCV, ECCV, AAAI, WACV, BMVC, ICASSP, IJCNN, TCSVT, Solar Energy
- Technical Committee, BMVC Privacy and Fairness Workshop 2024
- Graduate Teaching Assistant, Computer Logic and Organization (*COP 3103*), 2026
- Local SAT tutoring, 2025

Character Referees _____

- Dr. Mubarak Shah, UCF Board of Trustees Chair Professor, University of Central Florida, shah@crcv.ucf.edu
- Dr. Kristopher Davis, Professor, University of Central Florida, kristopher.davis@ucf.edu
- Dr. David Penn, Lecturer, University of Central Florida, david.penn@ucf.edu